



LikhaKahoy Woodcraft Shop

Week 11 - Final Project Phase 2 | NoSQL Reflection Essay

In the LikhaKahoy Woodcrafts e-commerce system, we mainly use MySQL for storing structured data such as customers, products, orders, and payments. It works well because the data is organized, and the relationships between tables are clearly defined, making it easier to manage and run queries.

However, in real systems, not all data is structured. Some types of data are more flexible and doesn't always fit neatly into tables. A good example is customer activity data, such as product views, searches, clicks, and browsing history. This kind of data changes frequently and does not follow a fixed format.

For this type of data, a NoSQL database like MongoDB is more suitable. Instead of using strict tables, it stores data in JSON-like documents, which makes it more flexibility. For instance, a single activity record can include the customer ID, the product viewed, the timestamp, and even the device used—all in one.

NoSQL databases are also helpful when handling large amounts of data because they can scale more easily and adapt to fast-changing information. This makes them useful for tracking user behavior and improving features like product recommendations. It also helps the system respond faster when dealing with real-time data, which is important for improving user experience in an e-commerce platform.

Another advantage is that NoSQL databases can handle unstructured or semi-structured data without needing constant changes to the database schema. This reduces the need for frequent redesigns when new types of data are introduced. As the system grows, this flexibility becomes more important because user behavior and business requirements can change over time..

Overall, MySQL is more suitable for structured data, while NoSQL databases are better for handling flexible or unstructured data. Using both together can make the system more efficient, scalable, and practical for real-world use.